Shishu-Sneh

Baby's First Year Guide

Android App Development Project Report

Infant Healthcare & Growth Monitoring – Android Application

Project Title	Android App Development using GenAI – Shishu-Sneh (Healthcare) – Baby's First
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1. Introduction

Shishu-Sneh is an Android application developed to support new mothers — particularly those in rural and semi-urban India — in tracking their infant's health, growth, and immunisation milestones during the critical first year of life. The application name combines the Sanskrit word *Shishu* (infant) with *Sneh* (affection / care), reflecting its core purpose: to be a compassionate, intelligent digital companion for every new mother.

Developed using Kotlin with XML-based layouts, the app integrates Room Database for permanent on-device storage, WorkManager for scheduling vaccination and milestone reminders, and MPAndroidChart for visual growth trend lines. The application targets the maternal and infant-care segment of India's healthcare ecosystem — a segment that has historically depended on paper health cards, fragmented hospital records, and oral community wisdom to guide infant care decisions.

The application addresses three critical needs of new mothers: tracking the infant's weight and height growth over time, receiving timely vaccination reminders calibrated to the baby's date of birth, and accessing weekly developmental milestone checklists that guide parents through the baby's first year. A soft, mother-friendly UI ensures that the app is accessible even to users with limited digital experience.

Shishu-Sneh is designed to function entirely offline, requiring no internet connection for its core operations.

Data is stored persistently on-device using Room Database, ensuring that health records survive app restarts, device reboots, and long periods of inactivity. The app targets the vast and underserved community of new mothers across Karnataka and rural India who lack access to paediatric clinics or digital health platforms.

2. Problem Statement

India records millions of infant births every year, yet access to consistent paediatric guidance remains limited for mothers in villages and small towns. New mothers often discontinue exclusive breastfeeding early due to myths, social pressure, or a return to work. Once they leave the hospital, there is no reliable system to track the baby's weight gain, vaccination schedule, or developmental progress. Several systemic problems affect this segment:

Challenge	Description
Paper Health Cards	Hospital-issued growth cards are easily lost, damaged, or left at home during clinic visits, causing gaps in the infant's health history.
No Vaccination Reminders	Parents miss critical immunisation windows — including Polio Drops and DPT boosters — because there is no automated reminder system tied to the baby's individual date of birth.
Myth-Driven Decisions	Without access to expert guidance in the local language, mothers follow community myths that may harm exclusive breastfeeding or delay introduction of complementary foods.
No Growth Monitoring	Families have no way to visualise whether their baby's weight gain trend is healthy, making it impossible to detect early signs of stunting or malnutrition.
No Milestone Tracking	Parents are unaware of age-appropriate developmental milestones (e.g., holding head up, smiling, rolling over), leading to delayed identification of developmental delays.

These interconnected challenges highlight the urgent need for a dedicated, offline-first infant-health companion that mothers can use with minimal technical knowledge, on low-cost Android smartphones, without internet connectivity.

3. Project Vision

Shishu-Sneh envisions a world where every new mother — regardless of education level, economic status, or geographic location — has access to a reliable, intelligent, and empathetic infant-health companion in her own language.

The project is built on four foundational pillars:

- **Accessibility –** Infant-health guidance must reach every mother, regardless of smartphone model, internet availability, or digital literacy. The app is designed for minimum API 21 devices and requires no internet for core features.

- **Simplicity –** The interface must be usable by mothers who may have limited literacy and no prior experience with health apps. Soft colours, large touch targets, clear iconography, and a minimal learning curve are non-negotiable design principles.
- **Reliability –** Health records must never be lost. Room Database provides ACID-compliant local storage that survives app crashes, device reboots, and accidental app closures.
- **Automation –** The burden of remembering vaccination dates and milestone checks must be removed through technology. WorkManager schedules reminders automatically based on the baby's date of birth, without any manual intervention from the mother.

Shishu-Sneh is not merely a note-taking app; it is a digital elder — a trusted companion that combines the wisdom of a community health worker with the precision of a paediatric schedule, available 24 hours a day in the palm of a mother's hand.

4. Scope of the Application

4.1 Target Users

- New mothers in rural and semi-urban India, particularly first-time parents.
- Mothers who have recently delivered and been discharged from government hospitals.
- Accredited Social Health Activists (ASHAs) and Anganwadi workers who monitor infant health.
- Any parent or caregiver responsible for tracking an infant's first-year development.

4.2 Platform

- Android smartphones with minimum API Level 21 (Android 5.0 Lollipop and above).
- Optimised for low-end and mid-range devices with limited RAM.
- APK size kept minimal to support devices with limited internal storage.

4.3 Offline Functionality

- All health and growth data is stored locally using Room Database — no cloud or server required.
- Core features including recording weight/height entries, viewing charts, and milestone checks work fullyoffline.
- Vaccination reminders use WorkManager, which requires no internet access.

4.4 Key Operational Scenarios

- Recording a new weight or height entry against the baby's age in months.
- Viewing a real-time growth trend line showing progress over time.
- Receiving automated vaccination reminders tied to the baby's date of birth.
- Completing a weekly milestone checklist and tracking which milestones are achieved.
- Reading a daily feeding guide with age-appropriate nutritional tips for mother and baby.

5. Objectives

The project is aligned with the following measurable objectives:

- **Provide a fast and accurate growth recording system** – Ensure that a new weight or height entry can be saved within one screen interaction from the home dashboard, with fields for age (months), weight (kg), and height (cm).
- **Deliver a visual growth trend line** – Render an MPAndroidChart line chart showing the baby's weight-for-age trajectory, updated instantly with every new entry, clearly showing progress or concern over 6+ months.
- **Ensure permanent, offline data persistence** – All health records must survive application restarts, device reboots, and extended periods of inactivity using Room Database.
- **Automate vaccination reminders** – Enable WorkManager to schedule push notifications for upcoming vaccinations — including Polio Drops, DPT, BCG, and Hepatitis B — calculated from the baby's date of birth.
- **Provide a weekly developmental milestone checklist** – Display age-appropriate milestones (e.g., 'Is the baby holding their head up?') as a Yes/No checklist, updated weekly based on the baby's age.

- **Offer a daily feeding guide** – Deliver daily nutritional tips for both mother and baby, covering breastfeeding guidance and age-appropriate complementary feeding.
- **Support a soft, mother-friendly UI** – Design the interface using pastel colours, rounded cards, and gentle typography to create a calm and reassuring experience for new mothers.

6. Features

6.1 Core Features

Feature	Description
Growth Chart	Enter baby's weight and height once a month. Room Database stores each entry with age (months) and date. MPAndroidChart renders a smooth trend line showing progress over time. WHO growth reference bands are overlaid for context.
Vaccination Alert	WorkManager schedules notifications for every standard vaccination in India's national immunisation schedule, calculated from the baby's date of birth. Each alert includes the vaccine name, due date, and the disease it prevents.
Feeding Guide	A daily tip card delivers evidence-based nutritional guidance for the mother (breastfeeding diet, hydration) and age-appropriate complementary feeding advice for the baby, rotating through a curated content library.
Milestone Checklist	Weekly checklists present age-specific developmental milestones as Yes/No questions (e.g., 'Is the baby smiling responsively?'). Completed milestones are stored in Room Database and can be reviewed at any time.
Baby Profile	Mothers save the baby's name, date of birth, and gender. The date of birth drives all automated scheduling — vaccination alerts and milestone checklists are calibrated to this single anchor date.
Persistent Storage	Room Database (SQLite) stores all growth entries and milestone responses permanently ondevice. Data is retained after app closure, device restart, or extended inactivity.
Daily Reminder Notification	Mothers can set a daily reminder time via Settings. AlarmManager triggers a gentle push notification encouraging them to log today's feeding session or check the milestone list.
Settings Screen	Centralised settings for toggling daily reminders, setting reminder time, enabling or disabling vaccination alerts, and managing the baby's profile.

6.2 Future Scope Features

- GenAI Health Assistant – Integrate a local LLM to answer mothers' questions about feeding, cryingpatterns, and fever management in the local language.

- Telemedicine Integration – Allow mothers to share growth charts and milestone reports directly with aremote paediatrician via a secure link.
- Multi-Language Support – Extend the interface to Kannada, Hindi, Tamil, and Telugu in addition toEnglish.
- Cloud Backup – Optional encrypted cloud sync via Firebase to protect records against device loss.
- Sibling Profiles – Support tracking of multiple children under a single parent account.
- Biometric Authentication – Replace PIN login with fingerprint authentication on supported devices.

7. Functional Requirements

The following functional requirements define the core behaviours the system must support:

FR-1: Baby Profile Setup

- The system shall display a profile setup screen on first launch, prompting the mother to enter the baby'sname, date of birth, and gender.
- Profile data shall be stored in SharedPreferences and persist across all sessions.
- The baby's date of birth shall be used to calculate the baby's current age in weeks and months for allscheduling purposes.
- The mother shall be able to update the baby's profile at any time via the Settings screen.

FR-2: Growth Recording

- The system shall provide input fields for the baby's age (months), weight (kg), and height (cm).
- The system shall validate that weight and height are positive numeric values before saving.
- On successful save, the system shall store the entry in Room Database with a formatted timestamp andrefresh the growth chart immediately.
- The system shall display a trend line showing all recorded data points using MPAndroidChart.

FR-3: Growth Chart Dashboard

- The system shall render a line chart with age on the X-axis and weight on the Y-axis.
- The chart must update in real time after every new entry or deletion.
- The system shall display the most recent weight and height values in summary cards on the dashboard.

FR-4: Vaccination Alerts

- The system shall maintain a pre-loaded immunisation schedule aligned with India's NationalImmunisation Schedule.
- WorkManager shall calculate due dates for each vaccine based on the baby's date of birth and schedulereminders accordingly.
- Each notification shall include the vaccine name, scheduled date, and the disease it prevents.
- The mother shall be able to mark a vaccine as administered, which removes the pending alert.

FR-5: Milestone Checklist

- The system shall display a milestone checklist appropriate to the baby's current age in weeks.
- Each milestone shall be presented as a Yes/No question (e.g., 'Is the baby holding their head up?').
- Responses shall be stored in Room Database with a timestamp and shall be reviewable in a milestonehistory view.
- The checklist shall update automatically as the baby ages into a new milestone window.

FR-6: Daily Reminder Notification

- The mother shall be able to set a daily reminder time using a time picker dialog in Settings.
- The system shall use AlarmManager to trigger a push notification at the chosen time every day.
- The notification shall contain the baby's name and a prompt to log today's feeding or check themilestone list.
- The mother shall be able to toggle the daily reminder on or off independently.

FR-7: Feeding Guide

- The system shall display a daily feeding tip card on the home dashboard.
- Tip content shall be age-appropriate, sourced from a curated offline content library.
- Tips shall rotate daily and cover both maternal nutrition (breastfeeding diet) and baby feeding(complementary foods).

8. Non-Functional Requirements

Category	Requirement	Acceptance Criterion
Performance	**Load Time**	Dashboard must fully render within 800 ms of launch on a mid-range Android device.
Performance	**Entry Save**	A new growth entry must be saved and reflected in the chart within 500 ms of tapping Save.
Usability	**UI Simplicity**	Any user must be able to record a growth entry within 3 taps from the home screen.
Usability	**Legibility**	All primary text must use a minimum font size of 13sp. Chart labels must use a minimum of 11sp.
Reliability	**Data Persistence**	100% of health records must be retained after app closure, device restart, and 30+ days of inactivity.
Reliability	**Crash Rate**	Application crash rate must not exceed 1% during standard recording and navigation flows.
Accessibility	**Touch Targets**	All interactive elements must have a minimum touch target size of 48x48dp per Material Design guidelines.
Compatibility	**Android Versions**	Application must support Android 5.0 (API 21) through Android 14 (API 34).
Security	**Data Privacy**	All health data remains on-device. No data is transmitted to external servers.
Offline Support	**Core Features**	All core features (recording, charting, checklists, notifications) must function without any network connection.

9. System Architecture

Shishu-Sneh follows a single-tier, client-only architecture. All data processing, storage, and business logic occur entirely on-device. There are no backend servers, cloud databases, or API dependencies required for the application's core operation.

9.1 Technology Stack

Layer	Technology	Purpose
Language	**Kotlin**	Primary development language; concise, null-safe, fully interoperable with Android SDK.
UI Framework	**XML Layouts + AppCompat**	Traditional View-based UI with LinearLayout, RecyclerView, and Material components for broad device compatibility.
Charting	**MPAndroidChart**	Renders smooth growth trend line charts with customisable axes, WHO reference bands, and touch interaction.
Local Database	**Room Database (SQLite)**	ACID-compliant ORM over SQLite. Provides permanent ondevice storage of growth entries and milestone responses.
Background Tasks	**WorkManager**	Schedules and executes vaccination reminder notifications reliably, surviving device reboots and app closures.
Notifications	**AlarmManager + NotificationManager**	Triggers daily reminder notifications at mother-specified times. Supports Android 8+ notification channels.
Session Storage	**SharedPreferences**	Stores baby profile (name, DOB, gender), app settings, and reminder preferences as lightweight key-value pairs.
Build System	**Gradle (Groovy DSL)**	Dependency management, build configuration, and APK generation including kotlin-kapt for Room annotation processing.
Min. SDK	**API Level 21**	Targets approximately 98% of active Android devices globally, including low-cost smartphones prevalent in rural India.

9.2 Architectural Overview

The application follows a simplified MVC (Model-View-Controller) pattern appropriate for a single-activity, XMLlayout Android app:

- **Model Layer** – The data package contains the Room @Entity (GrowthEntry, MilestoneResponse), @Dao (GrowthDao, MilestoneDao), and @Database (AppDatabase) classes. This layer owns all data access logic and migration handling.
- **View Layer** – XML layout files define the UI structure (activity_main.xml, activity_growth.xml, activity_vaccinations.xml, activity_milestones.xml, activity_settings.xml, list_item_growth.xml). Custom drawables provide the soft, pastel visual styling.
- **Controller Layer** – Activity classes (MainActivity, GrowthActivity, VaccinationActivity, MilestoneActivity, SettingsActivity) orchestrate user interaction, invoke database operations on background threads via Executors, and update the UI on the main thread via runOnUiThread.
- **Background Workers** – VaccinationReminderWorker (Worker) queries the immunisation schedule, computes due dates from the baby's DOB, and posts notifications via NotificationManager. ReminderReceiver (BroadcastReceiver) handles daily notification delivery.

9.3 Database Design

The Room database contains two primary tables:

Table: growth_entries

id	INTEGER	PRIMARY KEY, AUTO	Auto-generated unique identifier for each growth entry.
ageMonths	INTEGER	NOT NULL	Baby's age in completed months at the time of recording.
weightKg	REAL	NOT NULL	Baby's weight in kilograms (stored as Double).
heightCm	REAL	NOT NULL	Baby's height in centimetres (stored as Double).
timestamp	TEXT	NOT NULL	Formatted date and time of entry recording.

Table: milestone_responses

Column	Type	Constraint	Description
id	INTEGER	PRIMARY KEY, AUTO	Auto-generated unique identifier.
ageWeeks	INTEGER	NOT NULL	Baby's age in weeks at the time of the milestone check.

Column	Type	Constraint	Description

milestoneKey	TEXT	NOT NULL	Identifier for the specific milestone question.
achieved	INTEGER	NOT NULL	Boolean stored as 0 (No) or 1 (Yes).
timestamp	TEXT	NOT NULL	Date and time the milestone response was recorded.

10. User Flow

The application is designed to minimise friction in daily use scenarios for new mothers. The complete user journey from app launch to health management is described below:

#	Screen	User Action & System Response
	Splash / Profile Setup	App launches. If first use, mother enters baby's name, date of birth, and gender. On subsequent launches, the home dashboard is shown directly.
2	**Home Dashboard**	Mother sees the main dashboard with the baby's name, current age, today's feeding tip, and quick-access cards for Growth Chart, Vaccinations, and Milestones.
	Growth Entry	Mother navigates to the Growth Chart screen. Enters the baby's current weight (kg) and height (cm). Taps Save. The entry is stored in Room Database and the trend line updates immediately.
4	**View Growth Chart**	The chart screen displays a line graph with age (months) on the X-axis and weight on the Y-axis. Each recorded data point is plotted. The most recent weight and height appear in summary cards above the chart.
	Vaccination Schedule	Mother navigates to the Vaccinations screen. A list shows all upcoming vaccines with their due dates (calculated from the baby's DOB), the vaccine name, and the disease it prevents. Administered vaccines can be marked as done.
6	**Milestone Checklist**	Mother opens the Milestone screen. Age-appropriate Yes/No questions are displayed. She taps Yes or No for each. Responses are saved to Room Database with the date.
	Daily Reminder	At the mother's chosen time, a gentle push notification appears: 'Time to log [Baby's Name]'s feeding and growth today!' Tapping opens the dashboard.

8	**Vaccination Alert**	WorkManager fires when a vaccine is due. A notification appears: 'Polio Drops are due for [Baby's Name] today – Prevents Poliomyelitis.' No user action required.
	Settings	Mother navigates to Settings via the 3-dot menu. She can update the baby's profile, toggle daily reminders, set reminder time, and manage vaccination alerts.

11. Technical Implementation

11.1 Room Database & Migration

Growth entries and milestone responses are stored using the Room persistence library, which provides a typesafe abstraction over SQLite. The GrowthDao interface exposes four operations: getAll() (ordered by ageMonths ASC to display chronological growth trend), insert(), delete(), and deleteAll(). The AppDatabase singleton uses a double-checked locking pattern to ensure a single database instance throughout the app lifecycle. A database migration (version 1 → version 2) handles the addition of the milestone_responses table using an ALTER TABLE / CREATE TABLE statement, ensuring that existing users who upgrade do not lose their growth history.

11.2 MPAndroidChart Integration

The growth trend line is rendered using the MPAndroidChart library. Growth entries are converted to a list of Entry objects (x = ageMonths, y = weightKg) and supplied to a LineDataSet. The chart is configured with a smooth cubic bezier mode, custom axis labels, and a pastel fill gradient beneath the line to create a soft, visually reassuring appearance. WHO weight-for-age reference values are overlaid as a secondary LimitLine series, allowing mothers to immediately see whether their baby's growth is on track.

11.3 WorkManager for Vaccination Reminders

VaccinationReminderWorker extends Worker and implements doWork() to check the national immunisation schedule against the baby's date of birth stored in SharedPreferences. For each vaccine whose due date falls within the next 7 days, a push notification is posted via NotificationManager. The worker is enqueued as a PeriodicWorkRequest with a 1-day interval using WorkManager.enqueueUniquePeriodicWork() with ExistingPeriodicWorkPolicy.REPLACE, ensuring reliable daily checks.

11.4 AlarmManager for Daily Reminders

The daily notification reminder uses AlarmManager.setRepeating() with AlarmManager.RTC_WAKEUP and an interval of AlarmManager.INTERVAL_DAY. The alarm triggers ReminderReceiver, a BroadcastReceiver that builds and displays a NotificationCompat notification. On Android 8.0 and above, a notification channel is created before posting the notification. The PendingIntent includes FLAG_IMMUTABLE on API 23+ for security compliance.

11.5 Background Threading

Room does not permit database operations on the main (UI) thread. All database calls are executed using a single-threaded Executors.newSingleThreadExecutor() instance. Results are posted back to the UI thread via runOnUiThread{}, which updates the RecyclerView adapter and growth chart. This pattern avoids the need for LiveData or coroutines while maintaining thread safety.

11.6 Custom RecyclerView Adapter

GrowthAdapter extends RecyclerView.Adapter and inflates the custom list_item_growth.xml layout for each entry. The onBindViewHolder() method populates age (months), weight (kg), height (cm), and formatted timestamp. The adapter is backed directly by a mutable list in GrowthActivity, and notifyDataSetChanged() is called after every database operation.

11.7 Baby Profile & SharedPreferences

The baby's name, date of birth (stored as a long epoch timestamp), and gender are saved in SharedPreferences under a named file (BabyProfilePrefs). All scheduling calculations — vaccination due dates, milestone age windows, and daily reminder content — derive from the stored date of birth, ensuring that the entire app experience is personalised to the individual baby's age.

12. Project File Structure

File / Directory	Purpose
app/build.gradle	App-level Gradle config: plugins, SDK versions, Room, WorkManager, MPAndroidChart, and kapt dependencies.
data/GrowthEntry.kt	Room @Entity data class representing a single weight/height growth entry.
data/MilestoneResponse.kt	Room @Entity data class representing a single milestone Yes/No response.
data/GrowthDao.kt	Room @Dao interface with getAll, insert, delete, and deleteAll queries for growth entries.
data/MilestoneDao.kt	Room @Dao interface with queries for inserting, retrieving, and clearing milestone responses.
data/AppDatabase.kt	Room @Database singleton with version management and migration definitions for both tables.
MainActivity.kt	Home dashboard: feeding tip card, quick-access navigation cards, and baby profile header.
GrowthActivity.kt	Growth chart screen: entry form, MPAndroidChart trend line, and growth history list.
VaccinationActivity.kt	Vaccination schedule screen: list of upcoming and completed vaccines with due dates.
MilestoneActivity.kt	Milestone checklist screen: age-appropriate Yes/No questions and response history.

SettingsActivity.kt	Settings for daily/vaccination reminders, reminder time picker, and profile management.
ReminderReceiver.kt	BroadcastReceiver that builds and posts daily reminder push notifications.
VaccinationReminderWorker.kt	WorkManager Worker that checks immunisation schedule and dispatches vaccination alerts.
GrowthAdapter.kt	Custom RecyclerView.Adapter inflating list_item_growth.xml for each growth entry.
res/layout/activity_main.xml	Dashboard layout: baby header, feeding tip card, and navigation cards.
res/layout/activity_growth.xml	Growth chart layout: entry form, MPAndroidChart view, and history list.
res/layout/activity_vaccinations.xml	Vaccination schedule layout with RecyclerView of vaccine items.
res/layout/activity_milestones.xml	Milestone checklist layout with scrollable Yes/No question cards.
res/layout/activity_settings.xml	Settings layout with switches, time picker button, and profile form.
res/layout/list_item_growth.xml	Custom list row: age (months), weight, height, and timestamp.
res/drawable/*.xml	Shape drawables for card backgrounds, pastel gradients, and avatar circles.

File / Directory	Purpose
res/menu/main_menu.xml	3-dot overflow menu: Profile, Settings.

AndroidManifest.xml	App manifest: permissions, activity declarations, receiver registration launcher activity.

13. Permissions & Privacy

Permission	Required For	Notes
POST_NOTIFICATIONS	Daily reminders and vaccination alerts	Required on Android 13+. User is prompted on first notification trigger.
SCHEDULE_EXACT_ALARM	Daily reminder at exact time	Required on Android 12+. User may need to grant via system settings.
RECEIVE_BOOT_COMPLETED	Restart alarms after device reboot	Ensures daily reminders survive device restarts.

No personal user data is transmitted to any external server. All health records, baby profile information, and preference data remain exclusively on the mother's device. The application does not collect analytics, crash reports, or usage statistics. The baby's date of birth is used solely for scheduling vaccination reminders and milestone checklists, and is never shared outside the app.

14. Impact Goals

14.1 Child Survival

By delivering timely, personalised vaccination reminders calibrated to each baby's date of birth, Shishu-Sneh directly reduces the risk of vaccine-preventable infant mortality. Diseases such as Poliomyelitis, Pertussis, and Hepatitis B can be prevented through timely immunisation; missed vaccination windows are a leading cause of preventable infant deaths in rural India.

14.2 Stunting Prevention

The growth chart feature empowers mothers to track weight-for-age trends and compare them against WHO reference bands. Early identification of growth faltering allows families and healthcare workers to intervene before stunting becomes irreversible. The app transforms passive observation into an active, data-driven monitoring habit.

14.3 Maternal Empowerment

Daily feeding tips and breastfeeding guidance, delivered in accessible language, counter myths and social pressure that cause early discontinuation of exclusive breastfeeding. Informed mothers make better feeding decisions, directly improving their infant's nutritional outcomes during the critical 0-6 month window.

14.4 Health Equity

By operating fully offline and targeting minimum API 21 devices, Shishu-Sneh delivers expert paediatric guidance to mothers who have no access to private clinics, internet connectivity, or literate community health

resources. Digital health equity begins with apps that work on the cheapest smartphone in the most remote village.

Android App Development using GenAI

14.5 Scalability of Impact

The modular architecture of Shishu-Sneh — with separate data, activity, and worker layers — supports rapid expansion. Future versions can add multi-language support, GenAI-powered health assistants, telemedicine integration, and cloud backup without requiring a fundamental redesign of the existing codebase.

15. Success Criteria

The project will be considered successfully completed when the following criteria are met:

#	Success Criterion	Measurement	Priority
1	App loads and reaches the Dashboard within 800 ms on a reference device (Redmi 9A or equivalent).	Android Profiler measurement	Critical
2	Growth chart must show a clear progress trend over 6 months of data.	Manual chart validation with sample data	Critical

3	Vaccination reminder is triggered correctly based on the baby's date of birth.	Manual date-based scheduling test	Critical
4	All growth records are retained after device reboot and 30+ days of inactivity.	Room DB inspection test	Critical
5	Milestone checklist updates correctly when baby crosses a new age threshold.	Manual age progression test	Critical
6	Daily notification appears at the set time with correct baby name and content.	Manual notification test	High

7	Summary cards update correctly (latest weight and height) after every add and delete.	Manual validation	High
8	App does not crash during complete walkthrough of all screens and features.	Manual regression testing	High

9	Baby name and age appear correctly in the dashboard header and notifications.	Manual UI and notification check	**Medium**
10	All features work fully in Airplane Mode.	Airplane Mode test on device	**Medium**

Conclusion

Shishu-Sneh represents a focused and socially meaningful application of Android development in the domain of infant healthcare and maternal guidance. By addressing a clear and well-defined problem — the lack of a reliable, offline, and intelligent digital health companion for new mothers in rural India — the project delivers measurable practical value alongside technical merit.

The application's architecture prioritises reliability, simplicity, and offline-first operation above all else. Every design decision — from Room Database for permanent storage, to WorkManager for resilient vaccination scheduling, to MPAndroidChart for intuitive growth visualisation — is motivated by a single question: will this work for a new mother in a village with a basic smartphone and no internet connection?

The project also establishes a strong technical foundation for future expansion. The separation of data, view, and controller layers supports the addition of new features — GenAI health assistants, multi-language support, cloud backup, telemedicine integration — without requiring a fundamental redesign of the existing codebase. The Room migration framework ensures that future schema changes can be deployed to existing users without data loss.

In conclusion, Shishu-Sneh is not simply a technical deliverable — it is a small but meaningful step toward

healthier babies, more confident mothers, and a more equitable healthcare system for the millions of families who form the heart of India's future.